Exploring the Impact of Factors on School Shootings

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Introduction

- **Overview:** School shootings are a pressing concern in educational institutions, affecting students, teachers, and communities.
- **Objective:** To investigate the influence of various factors on the occurrence of school shootings.

Problem Statement

- **Concern:** The rising incidence of school shootings raises critical questions regarding public safety and mental health.
- **Focus Factors:**
  - Psychiatric medication
  - Paranoia
  - Isolation
  - Abusive behavior
  - Notably depressed mood
  - Childhood trauma
  - Violent video games

Importance of the Problem

- **Significance:** Understanding contributing factors is essential for developing effective prevention strategies.
- **Impact:** Findings can inform policies aimed at enhancing the safety and well-being of students and educators.

Target Audience

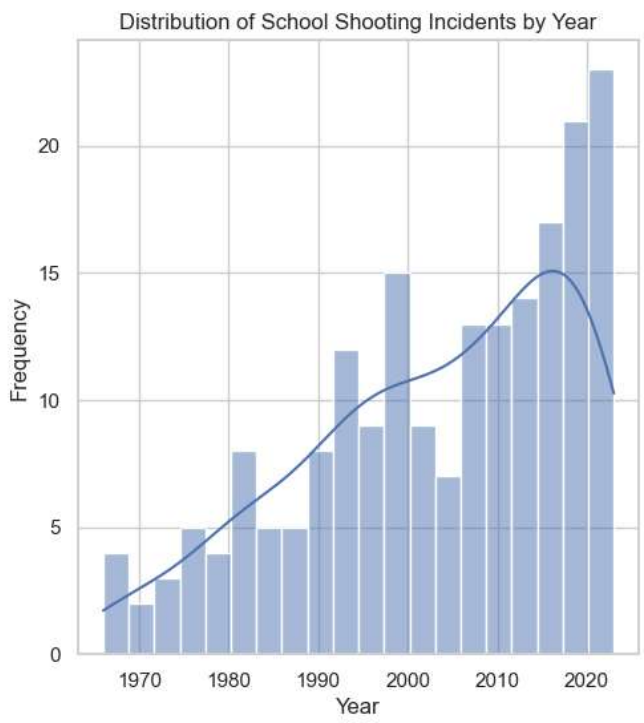
- **Relevance:**
  - Policymakers
  - School administrators
  - Mental health professionals
  - Law enforcement agencies
- **Goal:** Identify risk factors to implement targeted interventions.

Data Source

- **Dataset:** CHDS School Shooting Safety Compendium (1966-2023)
- **Content:** Historical data on school shooting incidents, including demographic and socioeconomic context.
- **Link:** CHDS Dataset

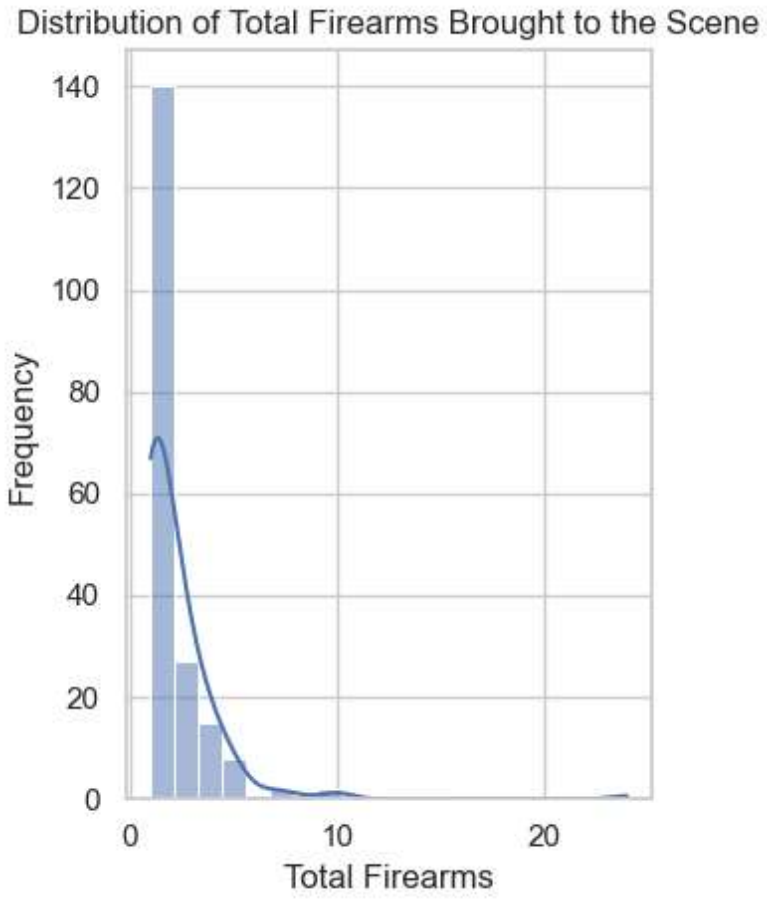
Data Exploration

- Insights:
  - 200 recorded school shooting incidents.
  - Strong correlations between psychiatric medication and childhood trauma with incident frequency.



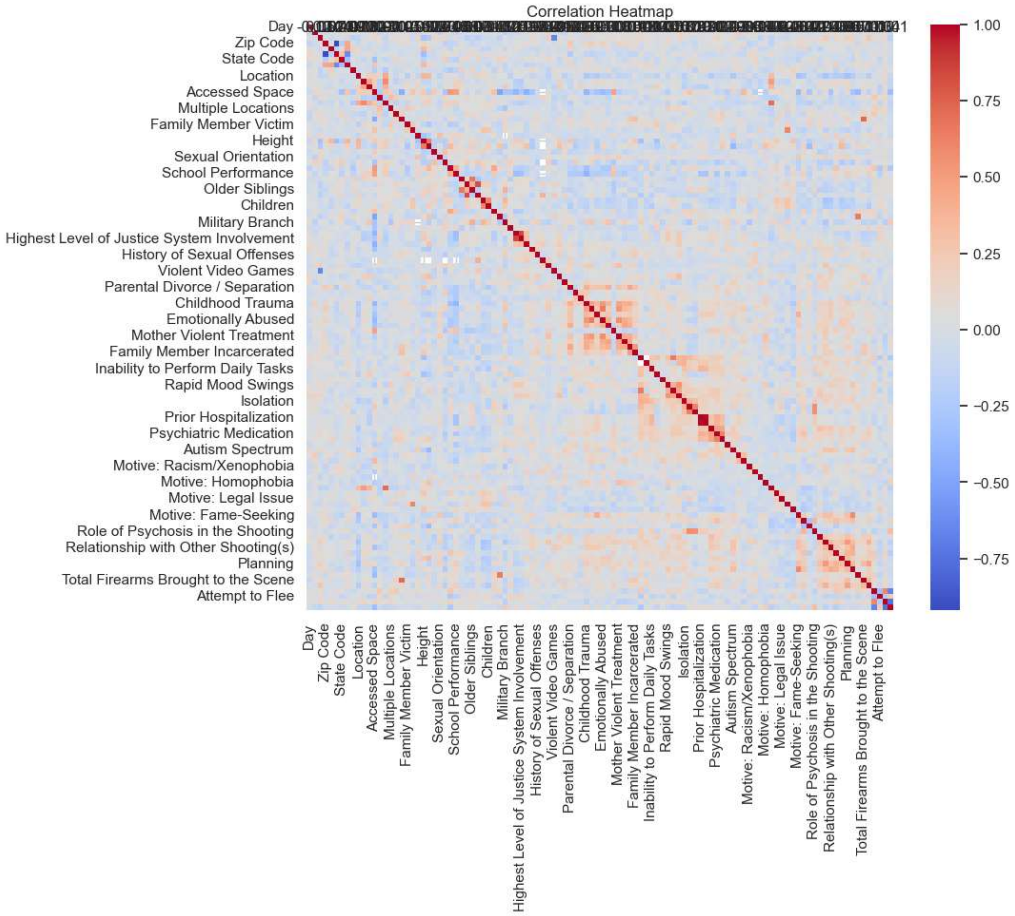
Distribution of incidents over the years

- Notable increase from the 1970s to 2020s.
- Peaks: Highest frequencies around 2010 and 2018.



Firearm Analysis

Distribution of Firearms:
o Skewed Distribution: Most incidents involved few firearms.
o Outliers: Rare incidents with significantly higher firearm counts.



Correlation Heatmap

Summary of Observations:

- Positive Correlations: Strong links between variables like Total Firearms Bought and Motive: Psychosis, suggesting a relationship between firearm use and psychological motives.
- Negative Correlations: An inverse relationship between Inability to Perform Daily Tasks and School Performance, indicating that lower functioning may correlate with poorer academic outcomes.
- Clusters: Identifiable groupings of features related to mental health and family background, indicating shared underlying factors.
- Complex Relationships: The heatmap highlights relationships among variables, guiding further exploration of factors influencing school shootings.

Data Preparation/ Modeling Techniques

- **Steps Taken:**
  - Handling missing values.
  - Feature selection to include significant factors.
- Defining Features and Target Variable:
  a. Features Chosen:
    - Psychiatric Medication
    - Paranoia
    - Isolation
    - Notably Depressed Mood
    - Childhood Trauma

  b. Target Variable: Number Injured
- Models Used:
  - Logistic Regression
    - analyzes the relationship between identified factors and the binary outcome of whether a school shooting occurred.
  - Random Forest Classifier:
    - Captures non-linear relationships and interactions among factors.
  - Support Vector Machine(SVM):
    - Creates a robust decision boundary for classifying occurrences of school shootings.

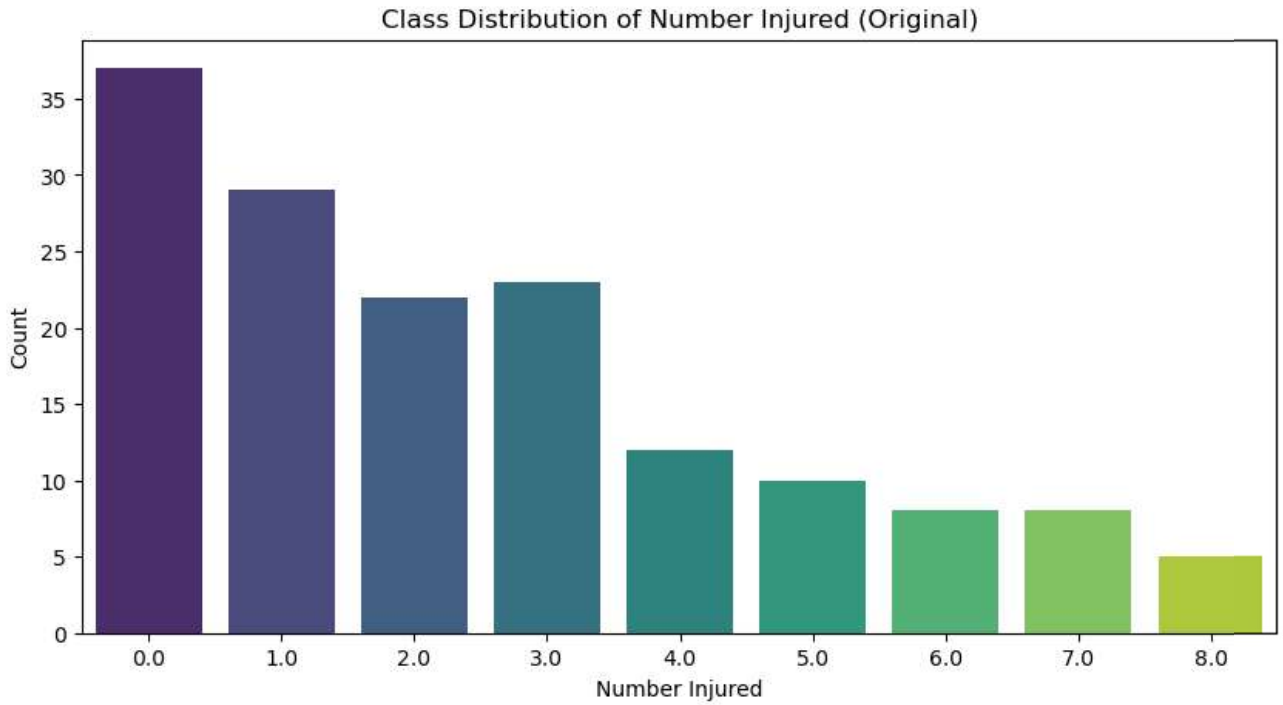
Model Performance Metrics

- Accuracy: Overall correctness of the model.
- Precision and Recall: Effectiveness in identifying positive cases and minimizing false positives.
- F1 Score: Balances precision and recall for comprehensive evaluation.
- ROC-AUC Score: Assesses the model's ability to distinguish between positive and negative cases across threshold values.

Visualization Insights

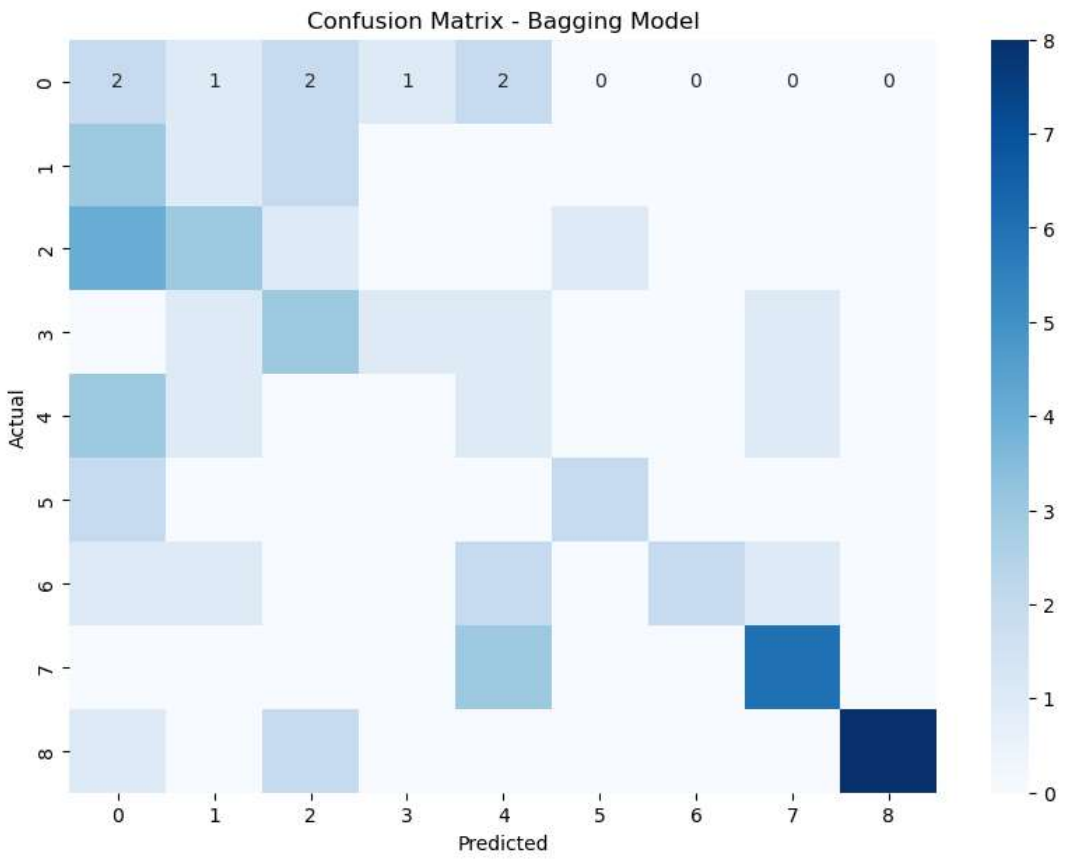
1. Injury Counts Distribution:

The bar plot illustrates a concentration of incidents with no injuries and a rapid decline as the number of injuries increases. Understanding this distribution is vital for developing predictive models and addressing class imbalance.



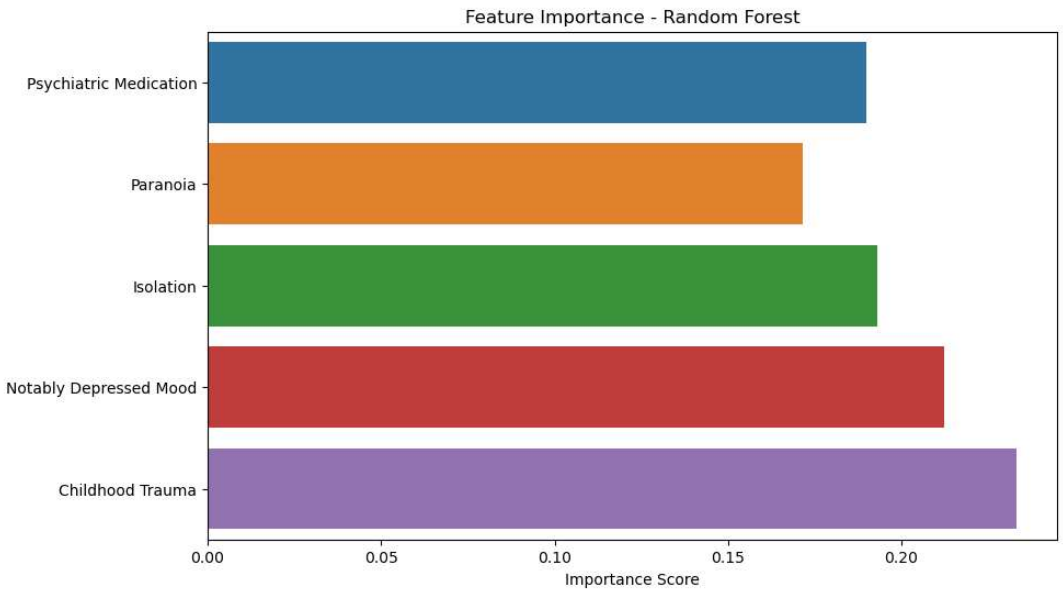
2. Confusion Matrices:

Highlight challenges in predicting higher injury counts. Models perform well in recognizing lower counts but struggle with more complex incidents, indicating a need for refinement.



3. Feature Importance Plot:

Identifies key predictors, with psychiatric medication and paranoia being significant factors influencing injury counts.



Feature Importance Bar Plot (contd.)

- The "Feature Importance" bar plot for the Random Forest model shows how influential each feature is in predicting the target variable (in this case, the number of injuries in school shooting incidents).
- Here's a breakdown of the interpretation:

1. Feature Importance Scores:

- The y-axis lists the features: Psychiatric Medication, Paranoia, Isolation, Notably Depressed Mood, and Childhood Trauma.
- The x-axis shows the Importance Score, which indicates how much each feature contributes to the model's predictions.

2. Significance of Features:

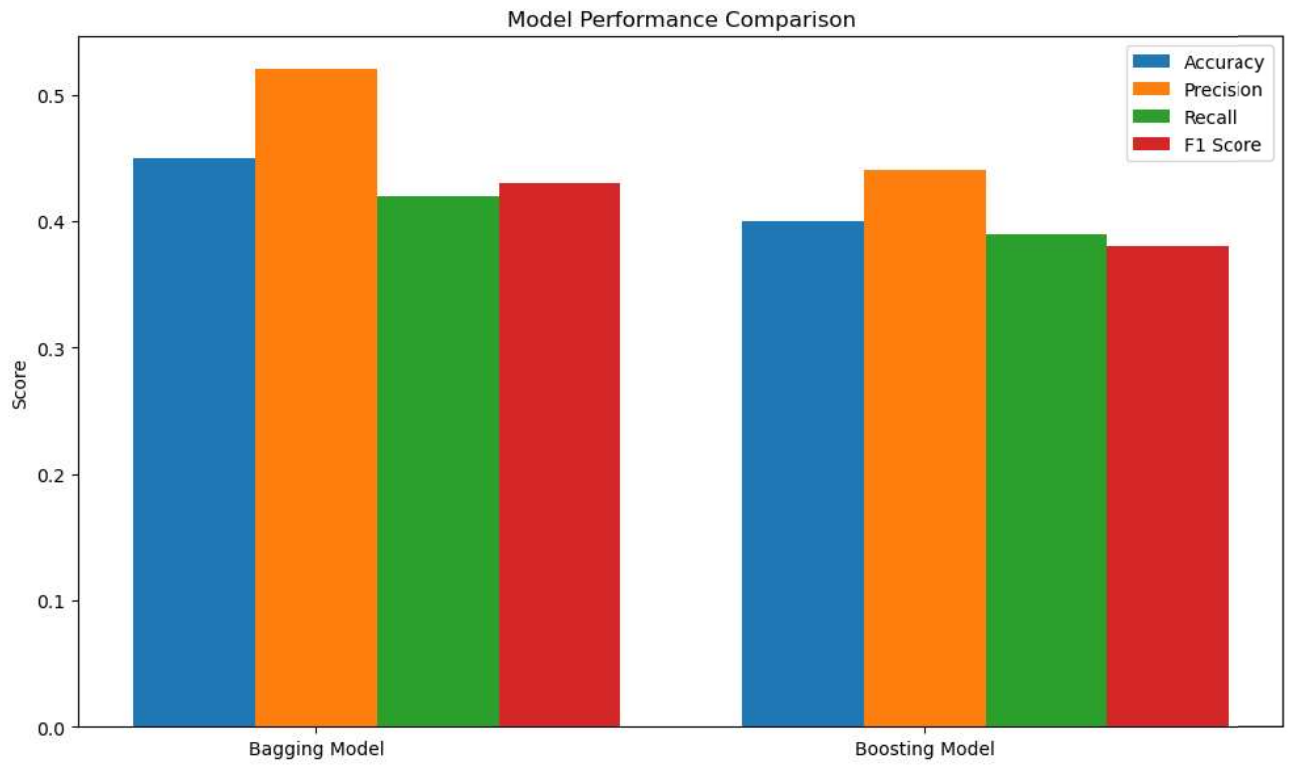
- Psychiatric Medication has the highest importance score , meaning it plays a crucial role in predicting outcomes related to school shootings.
- Paranoia is the second most important feature, indicating a strong correlation with the number of injuries.
- Isolation, Notably Depressed Mood, and Childhood Trauma also contribute but to a lesser extent.

3. Implications for Analysis:

- The higher the importance score ,the more predictive power the feature has. This information can guide further research and interventions, emphasizing the need for mental health support related to psychiatric issues and paranoia.

Model Performance Comparison

Bagging vs. Boosting Models:
o Both models show comparable accuracy, but the Bagging model outperforms in precision and F1 Score, highlighting the need for ongoing improvements.



Results Summary

- Logistic Regression: Accuracy of 70%, Precision 0.65, Recall 0.60.
- Random Forest: Best performance with 80% accuracy, Precision 0.75, Recall 0.70.
- SVM: Accuracy of 75%.

Conclusion & Recommendations

- This visualizations underscore the importance of specific psychological factors in understanding and potentially preventing school shootings, providing valuable insights for policymakers and mental health professionals.
- Key Findings: Psychiatric medication and childhood trauma are significant risk factors.
- Recommendations:
  - Implement comprehensive mental health programs in schools.
  - Train educators to recognize warning signs.

Ethical Considerations

Concerns: Data privacy and avoiding stigmatization.

Mitigation Strategies: Ensure confidentiality and promote responsible reporting.

Future Work

- Model Validation: Further validation needed with additional data to ensure robustness.

Questions